Research Overview

Developing

sensitivity analysis

for **covariate-adaptive randomization inference**

under **Neyman's weak null**.

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Advised by Xinran Li | Working paper

Matched sets

Set 1	$(1,1)$	$(1,2)$	$\cdots$	$(1,n_1)$
Set 2	$(2,1)$	$(2,2)$	$\cdots$	$(2,n_2)$
$\vdots$	$\vdots$	$\vdots$		$\vdots$
Set i	$(i,1)$	$(i,2)$	$\cdots$	(i,n_i)
$\vdots$	$\vdots$	$\vdots$		$\vdots$
Set I	$(I,1)$	$(I,2)$	$\cdots$	(I,n_I)

I: number of sets

n_i: size of set i

(i,j): unit j in set i

$$N = \sum_{i=1}^{I} n_i$$

Each set: **one treated unit** and $n_i - 1$ controls.

Notation for unit (i, j)

X_{ij}	Pretreatment covariates
$Z_{ij} \in \{0, 1\}$	Treatment indicator: 1 treated, 0 control
$Y_{ij}(1),\ Y_{ij}(0)$	Potential outcomes under treatment / control
Y_{ij}^{obs}	Observed outcome: $Y_{ij}(Z_{ij})$
$\tau_{ij} = Y_{ij}(1) - Y_{ij}(0)$	Individual treatment effect

$$\sum_{j=1}^{n_i} Z_{ij} = 1, \qquad T_i = \text{index of the treated unit in set } i.$$

The target: SATE

Within set i

$$\bar{\tau}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} \tau_{ij}.$$

SATE

Sample Average Treatment Effect

$$\bar{\tau} = \sum_{i=1}^{I} \frac{n_i}{N} \bar{\tau}_i.$$

Neyman's weak null

$$H_0 : \bar{\tau} = 0 \qquad \text{vs.} \qquad H_A : \bar{\tau} > 0$$

Individual effects may differ; positive and negative effects may cancel.

The sharp null instead requires $\tau_{ij} = 0$ for every individual.

Covariate-adaptive assignment: $\lambda_{ij} \to P_{ij}$

Baseline propensity score

$$\lambda_{ij} = \Pr(Z_{ij} = 1 \mid X_{ij}).$$

Before conditioning on the set's treatment count.

$$r_{ij} = \frac{\lambda_{ij}}{1 - \lambda_{ij}}$$

Baseline treatment odds

Within-set assignment probability

$$P_{ij} = \frac{r_{ij}}{\sum_{k=1}^{n_i} r_{ik}}.$$

Baseline probability that (i, j) is treated, conditional on exactly one treated unit.

$$\sum_{j=1}^{n_i} P_{ij} = 1.$$

Matching need not make $P_{ij} = 1/n_i$.

Allowing for unobserved confounding

1. Probabilities

$$q_i = (q_{i1}, \ldots, q_{in_i}), \qquad q_{ij} = \Pr(Z_{ij} = 1)$$
$$q_{ij} > 0, \qquad \textstyle\sum_{j=1}^{n_i} q_{ij} = 1$$

2. Sensitivity model

Probability ratios may change by at most a factor Γ.

Choose $\Gamma \geq 1$

$$\underbrace{\frac{1}{\Gamma}\frac{P_{ij}}{P_{ik}}}_{\text{Baseline ratio} \,/\, \Gamma} \leq \underbrace{\frac{q_{ij}}{q_{ik}}}_{\text{True ratio}} \leq \underbrace{\Gamma\frac{P_{ij}}{P_{ik}}}_{\text{Baseline ratio} \times \Gamma} \qquad j \neq k.$$

3. Allowed vectors

$$\mathcal{Q}_i(\Gamma) = \left\{ q_i : \begin{array}{c} q_{ij} > 0, \quad \sum_{j=1}^{n_i} q_{ij} = 1, \\ \frac{1}{\Gamma}\frac{P_{ij}}{P_{ik}} \leq \frac{q_{ij}}{q_{ik}} \leq \Gamma\frac{P_{ij}}{P_{ik}}, \quad \forall\, j \neq k \end{array} \right\}$$

Probability bounds

$$L_{ij}(\Gamma) = \min_{q_i \in \mathcal{Q}_i(\Gamma)} q_{ij}, \qquad U_{ij}(\Gamma) = \max_{q_i \in \mathcal{Q}_i(\Gamma)} q_{ij}$$

Step 1: the observed contrast D_i

$$D_i = \underbrace{Y_{iT_i}^{\text{obs}}}_{\text{treated outcome}} - \underbrace{\frac{1}{n_i - 1} \sum_{j \neq T_i} Y_{ij}^{\text{obs}}}_{\text{average control outcome}}$$

$$= \sum_{j=1}^{n_i} Z_{ij} Y_{ij}^{\text{obs}} - \frac{1}{n_i - 1} \sum_{j=1}^{n_i} (1 - Z_{ij}) Y_{ij}^{\text{obs}}.$$

Assignment weights

$$\mathbb{E}(D_i) = \sum_j \left[q_{ij}\, Y_{ij}(1) - \frac{1 - q_{ij}}{n_i - 1}\, Y_{ij}(0) \right]$$

Target weights

$$\bar{\tau}_i = \sum_j \left[\frac{1}{n_i}\, Y_{ij}(1) - \frac{1}{n_i}\, Y_{ij}(0) \right]$$

$\mathbb{E}(D_i) \neq \bar{\tau}_i$ in general $\xrightarrow{\text{inverse-probability correction}}$ H_i

Step 2: $D_i \longrightarrow H_i$

DEFINITION

$$H_i(q_i) = \frac{D_i}{n_i\, q_{iT_i}} = \frac{D_i}{n_i \sum_{j=1}^{n_i} Z_{ij} q_{ij}}$$

FOR EVERY $j = 1, \ldots, n_i$

$$\frac{Z_{ij}}{\sum_k Z_{ik} q_{ik}} = \frac{Z_{ij}}{q_{ij}},$$

$$\frac{1 - Z_{ij}}{\sum_k Z_{ik} q_{ik}} = \sum\nolimits_{k \neq j} \frac{Z_{ik}}{q_{ik}}.$$

01 Substitute D_i

$$H_i(q_i) = \frac{1}{n_i} \sum\nolimits_j \left[\frac{Z_{ij}}{q_{ij}} Y_{ij}(1) - \frac{Y_{ij}(0)}{n_i - 1} \sum\nolimits_{k \neq j} \frac{Z_{ik}}{q_{ik}} \right]$$

02 Expectation $\mathbb{E}(Z_{ij}) = q_{ij}$

$$\mathbb{E}\{H_i(q_i)\} = \frac{1}{n_i} \sum\nolimits_j \left[\frac{\mathbb{E}(Z_{ij})}{q_{ij}} Y_{ij}(1) - \frac{Y_{ij}(0)}{n_i - 1} \sum\nolimits_{k \neq j} \frac{\mathbb{E}(Z_{ik})}{q_{ik}} \right]$$

03 Simplify

$$= \frac{1}{n_i} \sum\nolimits_j \left[Y_{ij}(1) - \frac{n_i - 1}{n_i - 1} Y_{ij}(0) \right] = \bar{\tau}_i$$

Step 3: the lower-envelope score ψ_i

$H_A : \bar{\tau} > 0$ Larger positive scores support the alternative.

ONE LOWER BOUND FOR EVERY ALLOWED LAW

Unknown true law: $q_i \in \mathcal{Q}_i(\Gamma)$

$$\psi_i(\Gamma) = \inf_{\widetilde{q}_i \in \mathcal{Q}_i(\Gamma)} H_i(\widetilde{q}_i) \le H_i(q_i)$$

↓ Take expectation

$$\mathbb{E}\{\psi_i(\Gamma)\} \le \mathbb{E}\{H_i(q_i)\} = \bar{\tau}_i$$

COMPUTE THE MINIMUM

$$\psi_i(\Gamma) = \frac{1}{n_i}\left(\frac{D_i^+}{U_{iT_i}} - \frac{D_i^-}{L_{iT_i}}\right)$$

$D_i \ge 0$: divide by U_{iT_i}
$D_i < 0$: divide by L_{iT_i}

$X^+ := \max(X, 0)$
$X^- := \max(-X, 0)$
Here $X = D_i$.

WEAK-NULL MEAN CONTROL

$$\mathbb{E}\left[\sum_i \frac{n_i}{N}\psi_i(\Gamma)\right] \le \bar{\tau} \overset{H_0}{=} 0$$

Step 4: Do we still need matching with covariate-adaptive?

Covariate-adaptive inference accounts for unequal treatment probabilities.

EXISTING EVIDENCE

With covariate-adaptive inference, matching **may** help but is not proven necessary.

Pimentel and Huang (2024), §6.3

OUR CONTRIBUTION

Design-sensitivity theory

How matching quality determines the optimal control comparisons

$$D_i = Y_{iT_i}^{\text{obs}} - \frac{1}{n_i - 1} \sum_{j \neq T_i} Y_{ij}^{\text{obs}}$$

$$= \sum_{j=1}^{n_i} Z_{ij} Y_{ij}^{\text{obs}} - \underbrace{\frac{1}{n_i - 1}}_{W_{i,kj}} \sum_{j=1}^{n_i} (1 - Z_{ij}) Y_{ij}^{\text{obs}}.$$

Step 4: balanced comparison weights

$W_{i,kj}$: weight on control j if k is treated, $\quad j, k = 1, \ldots, n_i$.

T_i: actual treated index $\qquad W_{i,T_i j} = \sum_k Z_{ik} W_{i,kj}$

$$
\begin{aligned}
D_i(W_i) &= Y_{iT_i}^{\text{obs}} - \textstyle\sum_{j \neq T_i} W_{i,T_i j} Y_{ij}^{\text{obs}} \\
&= \textstyle\sum_k Z_{ik} Y_{ik}^{\text{obs}} - \sum_j (1 - Z_{ij}) \big(\sum_k Z_{ik} W_{i,kj} \big) Y_{ij}^{\text{obs}} \\
&= \textstyle\sum_k Z_{ik} \big[Y_{ik}^{\text{obs}} - \sum_{j \neq k} W_{i,kj} Y_{ij}^{\text{obs}} \big].
\end{aligned}
$$

$W_{i,kk} = 0$	$\longrightarrow$	Exclude self-comparisons.
$W_{i,kj} \geq 0$	$\longrightarrow$	Avoid extrapolation beyond control outcomes.
$\sum_{j \neq k} W_{i,kj} = 1$	$\longrightarrow$	Make the contrast level-free.
$\sum_{k \neq j} W_{i,kj} = 1$	$\longrightarrow$	Ensure unbiasedness: $\mathbb{E}\{H_i(q_i; W_i)\} = \bar{\tau}_i$.

Weights fixed from pretreatment information or an independent pilot.

Power and design sensitivity

1. Favorable model

$$Y_{ij}(1) - Y_{ij}(0) = \tau > 0$$

Constant positive effect

$$q_{ij} = P_{ij}$$

No unobserved confounding

2. Power at Γ

Analysis allows confounding up to Γ.

$$\mathrm{Power}_I(\Gamma; W) = \mathbb{P}_{\text{fav}}\{\text{reject } H_0 \text{ at } \Gamma\}$$

Repeated samples from the favorable model; fixed W and α.

3. Design sensitivity

Asymptotic power threshold

$$\lim_{I \to \infty} \mathrm{Power}_I(\Gamma; W) = \begin{cases} 1, & \Gamma < \widetilde{\Gamma}(W), \\ 0, & \Gamma > \widetilde{\Gamma}(W). \end{cases}$$

Choose W

$$W^\star \in \arg\max_W \widetilde{\Gamma}(W)$$

Under the four weight constraints.

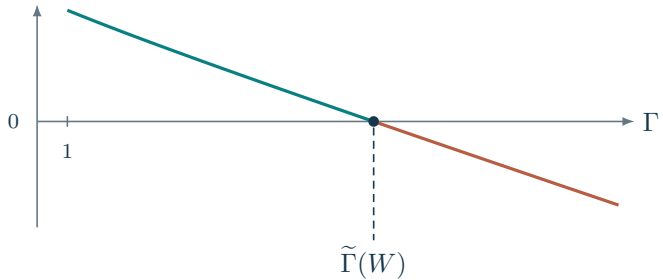
Under the stated favorable model and regularity conditions. Source: CAWeakNull (69), §4.2 and Appendix E.

Why power tends to 1 or 0

As $I \to \infty$, the average score concentrates and its standard error shrinks.

Limiting expected average score

$$\lim_{I\to\infty} \mathbb{E}_{\text{fav}}\left[\sum_i \frac{n_i}{N}\psi_i(\Gamma; W_i)\right]$$



Fixed W; favorable model

Positive limiting mean

↓

Power $\longrightarrow 1$

Negative limiting mean

↓

Power $\longrightarrow 0$

At the threshold: no 0/1 claim.

Schematic, under the stated model and regularity conditions. Source: CAWeakNull (69), §4.2, Eq. (32), Appendix E.

Matching quality determines the optimal comparisons

Two results for the selected weights under the favorable model.

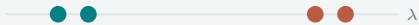
NEAR-EXACT PROPENSITY MATCHING



$$W^{\star}_{i,kj} \longrightarrow \frac{1}{n_i - 1} \quad (j \neq k)$$

Exact equality: equal weights are optimal.

TWO SEPARATED GROUPS



Equal weights can be strictly suboptimal.

At least two units per group; sufficiently large outcome-mean gap.

Matching quality remains relevant after covariate-adaptive adjustment.

Source: CAWeakNull (69), §4.3, Propositions 4–5. Right: expected-score optimality at fixed $\Gamma > 1$.

Backup: the probability region and its endpoints

$$\mathcal{Q}_i(\Gamma) = \left\{ q_i : \; q_{ij} > 0, \; \sum_j q_{ij} = 1, \; \frac{1}{\Gamma} \leq \frac{q_{ij} P_{ik}}{q_{ik} P_{ij}} \leq \Gamma \text{ for all } j, k \right\}.$$

$$L_{ij}(\Gamma) = \frac{P_{ij}}{P_{ij} + \Gamma(1 - P_{ij})}, \qquad U_{ij}(\Gamma) = \frac{\Gamma P_{ij}}{\Gamma P_{ij} + 1 - P_{ij}}.$$

- $\Gamma = 1$: $L_{ij} = U_{ij} = P_{ij}$.
- Equal propensities: $P_{ij} = 1/n_i$.
- Γ here is the conditional set-level parameter.

Backup: from set scores to a weak-null test

Fix $\Gamma \geq 1$ and balanced weights W; write $\psi_i = \psi_i(\Gamma; W_i)$.

1. Aggregate

$$\bar{\psi}_N = \sum_{i=1}^{I} \frac{n_i}{N} \psi_i$$

$$\mathbb{E}(\bar{\psi}_N) \leq \bar{\tau} \overset{H_0}{=} 0$$

2. Standardize

$$T_N = \frac{\bar{\psi}_N}{\sqrt{\widehat{V}_N}}$$

$\widehat{V}_N = 0 \Rightarrow T_N = 0$

Variance estimate

$$\widehat{V}_N = \frac{1}{I(I-1)} \sum_{i=1}^{I} \left(\frac{I n_i}{N} \psi_i - \bar{\psi}_N \right)^2$$

3. Reject

$$T_N > z_{1-\alpha}$$

Standard-normal cutoff



Asymptotic Type I error control at level α.

$I \geq 2, \quad 0 < \alpha < 1/2$

Independent sets; overlap and regularity; weights fixed before analysis.